HANNAH K. BAKO

Website: hannahbako.com ◇ Email: hbako@virginia.edu

ACADEMIC APPOINTMENTS

Assistant Professor — **University of Virginia**
School of Data Science — *Aug 2025 - Present*

PROFESSIONAL EXPERIENCE

User Research Intern — **Tableau**
User Research: Product Intelligence — *Summer 2022*

Design and Visualization Intern — **Apple**
AI/ML Visualization Team — *Summer 2021*

Android UX Designer — **N-Hub**
Jan 2017 - Jul 2017

EDUCATION

University of Maryland — **College Park, MD**
PhD. in Computer Science — *Aug 2025*

Stevens Institute of Technology — **Hoboken, NJ**
M.Sc. in Software Engineering — *May 2019*

Babcock University — **Ogun, Nigeria**
B.Sc. in Computer Information systems — *June 2015*

AWARDS AND RECOGNITION

Fellowships

UMD Summer Research Fellowship ($5000) — *2023*
NSF Travel Grant ($2850) — *2023*
CRA-WP Grad Cohort for Women Fellowship — *2022*
Dean's Fellowship, University of Maryland ($2500/yr) — *2019, 2020*

Service

Special recognition for outstanding reviews — *ACM CHI 2023*
Special recognition for outstanding reviews — *IEEE VIS 2023*
Eagle Leadership Award, Babcock University — *2015*

TEACHING EXPERIENCE

Instructor

DS 2003: Communicating with Data — Spring 2026

PUBLICATIONS

* Proceedings of IEEE VIS are published as a special issue of the flagship journal IEEE Transactions on Visualization and Computer Graphics (TVCG).

Journal Papers

J3 **VisAnatomy: An SVG Chart Corpus with Fine-Grained Semantic Labels** doi
Chen Chen, *H.K. Bako*, Peihong Yu, John Hooker, Simon C. Wang, Samuel Kim, Jessica Wu, Aouxe Ding, Lara Sandeep, Alex Chen, Chayanika Sinha, Zhicheng Liu
IEEE Transactions on Visualization and Computer Graphics
Proceedings of IEEE Visualization and Visual Analytics Conference (VIS '24).

J2 **Unveiling how Examples Shape Data Visualization Design Outcomes** doi
H.K. Bako, Xinyi Liu, Grace Ko, Hyemi Song, Leilani Battle and Zhicheng Liu
IEEE Transactions on Visualization and Computer Graphics
Proceedings of IEEE Visualization and Visual Analytics Conference (VIS '24).

J1 **Understanding how Designers Find and Use Data Visualization Examples** doi
H.K. Bako, Xinyi Liu, Leilani Battle, and Zhicheng Liu
IEEE Transactions on Visualization and Computer Graphics
Proceedings of IEEE Visualization and Visual Analytics Conference (VIS '22).

Conference Papers

C4 **How Do Observable Users Decompose D3 Code? A Qualitative Study** doi
Melissa Lin, Heer Patel, Medina Lamkin, *H.K. Bako*, Yuanjie Tu, Soham Raut, Leilani Battle
Proceedings of IEEE Visualization and Visual Analytics Conference (VIS '25)

C3 **Evaluating the Semantic Profiling Abilities of LLMs for NL Utterances in Data Visualization** doi
H.K. Bako, Arshnoor Bhutani, Xinyi Liu, Kwesi Cobinna, and Zhicheng Liu
Proceedings of IEEE Visualization and Visual Analytics Conference (VIS '24)

C2 **User-Driven Support for Visualization Prototyping in D3** doi
H.K. Bako, Alisha Varma, Anuoluwapo Faboro, Mahreen Haider, Favour Nerrise, Bissaka Kenah, John P Dickerson, and Leilani Battle
In Proceedings of the 28th International Conference on Intelligent User Interfaces (IUI '23)

C1 **Streamlining Visualization Authoring in D3 Through User-Driven Templates** doi
H.K. Bako, Alisha Varma, Anuoluwapo Faboro, Mahreen Haider, Favour Nerrise, Bissaka Kenah, and Leilani Battle
Proceedings of IEEE Visualization and Visual Analytics Conference (VIS '22)

Lightly Reviewed Papers and Articles

L1 **Tweets and Social Network Data for Twitter Bot Analysis** link
Jennifer Golbeck, Niloofarsadat Alavi, *H.K. Bako*, Saptarashmi Bandyopadhyay, Calvin Bao, et al.
Proceedings of SBP-BRiMS 2021: International Conference on Social Computing, Behavioral-Cultural Modeling and Prediction and Behavior Representation in Modeling and Simulation.

INVITED TALKS

Bridging Data and Example-Aided Ideation: A Journey Through Data Visualization Design

Worcester Polytechnic Institute, *Department of Computer Science*	Feb. 2025
University of Maryland Baltimore County, *Department of Information Systems*	Feb. 2025
University of Virginia, *School of Data Science*	Feb. 2025
Tufts University, *Department of Computer Science*	Mar. 2025
University of North Carolina Charlotte, *Department of Computer Science*	Mar. 2025
Washington University in St. Louis, *Department of Computer Science*	Mar. 2025

Innovation or Imitation? Measuring the influence of examples on visualization design
University of Maryland Human Computer Interaction Lab, *Lightening Talk* Oct. 2021

PROFESSIONAL SERVICE

Program Committee

IEEE Visualization and Visual Analytics (VIS) 2025

ACM Intelligent User Interfaces (IUI) 2025

Reviews

IEEE Transactions on Visualization and Computer Graphics (TVCG) 2025

IEEE Visualization and Visual Analytics (VIS) 2023-2025

ACM Conference on Human Factors in Computing Systems (CHI) 2023-2025

IEEE Pacific Visualization Conference (PacificVIS) 2024

ACM Intelligent User Interfaces (IUI) 2024

ACM Creativity & Cognition 2022

Service

ACM Intelligent User Interfaces Conference (IUI), *student volunteer* 2023

IEEE Visualization and Visual Analytics (VIS), *student volunteer* 2021

Denson Summer Enrinchment Program, *volunteer trainer* 2020

ACM Very Large Databases Confrence (VLDB), *student volunteer* 2020

Conference on Systems Engineering (CSER), *student volunteer* 2019

African Students Association (Stevens Institute of Technology), *founding member* 2019

Graduating Class Student Representatives (Babcock University), *Financial Secretary* 2015